Land Use / Land Cover (LULC) Classification

Using Multi-Seasonal Sentinel-2 Imagery and Ensemble Machine Learning

Final Project Report

Consolidating Experiments 1–4 and the Final Tuned Model

Executive Summary

This project developed a Land Use / Land Cover (LULC) classification system using multi-seasonal Sentinel-2 multispectral imagery and ensemble machine learning models. Four progressive experiments were conducted to evaluate the effects of feature engineering, temporal (seasonal) information, dataset size, and spectral index augmentation on classification performance, using Random Forest and XGBoost as the primary classifiers.

Across all experiments, XGBoost consistently outperformed Random Forest, and multi-seasonal spectral information provided a substantial and consistent improvement over single-season representations. Increasing dataset size alone, without richer features, did not meaningfully improve performance, whereas combining a large balanced dataset with four-season raw spectral bands produced the strongest results.

The final model — a hyperparameter-tuned XGBoost classifier trained on 40 raw multi-seasonal Sentinel-2 spectral features from a balanced 90,000-sample dataset — achieved the best overall performance of the entire project, with approximately 89% accuracy and a Macro F1-score of 0.89 on 18,000 unseen test samples.

Project Snapshot

Experiment	Description	Features	Samples	Classes	RF Accuracy	XGB Accuracy
EXP-01	Baseline — Raw multispectral bands (4 seasons)	40	27,000	9	85.00%	87.02%
EXP-02	Spectral indices + temporal features (3 seasons)	55	20,963	7	87.00%	88.00%
EXP-03	Increased sample size, raw bands (1 season)	10	90,000	9	85.00%	84.74%
EXP-04	Multi-season + increased sampling (± indices)	40 / 68	90,000	9	87.00%	88.21%
Final Model	**Tuned XGBoost — 40 multi-seasonal raw bands**	**40**	**90,000**	**9**	**—**	**~89.00%**

1. Introduction

1.1 Background

Land Use / Land Cover (LULC) classification is a fundamental task in remote sensing, supporting applications such as environmental monitoring, urban planning, agricultural assessment, and natural resource management. Satellite-based multispectral imagery, such as that provided by the Sentinel-2 mission, offers a cost-effective and scalable source of spectral information for distinguishing between different land cover types.

1.2 Project Objective

The objective of this project was to design and evaluate a robust LULC classification pipeline using Sentinel-2 multispectral data. Specifically, the project aimed to:

- Establish a baseline classification performance using raw spectral bands.
- Evaluate the impact of engineered features, including spectral indices and temporal difference features.

- Investigate the effect of dataset size on classification performance.
- Assess the benefit of incorporating multi-seasonal (temporal) spectral information.
- Identify and tune the best-performing model configuration to produce a final, deployable classifier.

1.3 Data Source

All experiments used Sentinel-2 multispectral satellite imagery, sampled across ten spectral bands (B2, B3, B4, B5, B6, B7, B8, B8A, B11, B12) and, where applicable, across up to four seasonal observation periods (Q1–Q4). Nine LULC classes were used across most experiments, labeled 10, 20, 30, 40, 50, 60, 80, 90, and 95.

1.4 Models Evaluated

Two ensemble tree-based classifiers were used consistently throughout the project:

- Random Forest (RF) — a bagging-based ensemble of decision trees.
- XGBoost (XGB) — a gradient-boosted ensemble of decision trees.

Both models were evaluated using accuracy, macro precision, macro recall, macro F1-score, and weighted F1-score, ensuring a balanced assessment across all classes.

2. Methodology Overview

The project followed an incremental, experimental approach: each experiment introduced one additional variable — feature engineering, sample size, or temporal coverage — while holding other factors as constant as possible, to isolate its effect on model performance.

Stage	Variable Investigated
Experiment 1	Baseline performance using raw multi-seasonal spectral bands
Experiment 2	Effect of spectral indices and temporal difference features
Experiment 3	Effect of increased sample size (single season)
Experiment 4	Combined effect of multi-season data, larger sampling, and spectral indices
Final Model	Hyperparameter tuning of the best configuration identified

3. Summary of Experiments

3.1 Experiment 1 — Baseline Using Raw Multispectral Bands

Established a baseline using 40 raw Sentinel-2 features (10 bands × 4 seasons) with no spectral indices, on a dataset of 27,000 samples across 9 classes. XGBoost outperformed Random Forest.

Model	Accuracy	Macro F1
Random Forest	85.00%	0.85
XGBoost	87.02%	0.87

Key takeaway: raw multi-seasonal bands alone provide a strong discriminative baseline, and XGBoost consistently edges out Random Forest even without feature engineering.

3.2 Experiment 2 — Spectral Indices and Temporal Feature Engineering

Combined 30 raw bands (3 seasons) with 21 spectral indices (BSI, EVI, NDBI, NDMI, NDVI, NDWI, SAVI) and 4 temporal difference features, for 55 total features, on a balanced dataset of 20,963 samples across 7 classes.

Model	Accuracy	Macro F1
Random Forest	87.00%	0.87
XGBoost	88.00%	0.88

Key takeaway: engineered features (indices + temporal differences) improved performance over the raw-band baseline, with XGBoost again the stronger model.

3.3 Experiment 3 — Effect of Increased Sample Size

Tested whether increasing sample size alone (10,000 samples/class, 90,000 total) could improve performance using only 10 raw single-season bands, with no additional feature engineering.

Model	Accuracy	Macro F1
Random Forest	85.00%	0.85
XGBoost	84.74%	0.85

Key takeaway: increasing sample size alone, without richer features or temporal information, did not improve accuracy — both models plateaued at ~85%, confirming that feature richness matters more than data volume alone.

3.4 Experiment 4 — Multi-Seasonal Features with Increased Sampling

Combined four-season raw bands (40 features) with the larger 90,000-sample dataset, and additionally tested Random Forest with spectral indices added (68 features).

Model	Configuration	Accuracy	Macro F1
Random Forest	40 Raw Bands	87.00%	0.87
XGBoost	40 Raw Bands	88.21%	0.88
Random Forest	40 Raw Bands + 28 Indices	87.00%	0.87

Key takeaway: this was the best-performing experimental configuration prior to tuning — XGBoost with 40 multi-seasonal raw bands achieved 88.21% accuracy. Adding spectral indices did not further improve Random Forest's overall performance.

4. Cross-Experiment Comparison

The table below traces the best accuracy achieved at each stage of the project, showing the cumulative effect of each experimental variable.

Stage	Key Change	Best Accuracy
EXP-01	Baseline (4-season raw bands, 27K samples)	87.02%
EXP-02	+ Spectral indices & temporal features (3-season)	88.00%
EXP-03	Single season, 90K samples (sampling test)	84.74%

Stage	Key Change	Best Accuracy
EXP-04	4-season raw bands, 90K samples	88.21%
Final Model	**EXP-04 configuration + hyperparameter tuning**	**~89.00%**

4.1 What Improved Performance

- Multi-seasonal information (1 season → 4 seasons): the single strongest driver of improvement, raising XGBoost accuracy from 84.74% (EXP-03) to 88.21% (EXP-04) — a gain of +3.47 points on an otherwise identical raw-band setup.
- Spectral indices and temporal-difference features (EXP-02): improved accuracy over the raw-band baseline within a smaller, feature-engineered configuration.
- Hyperparameter tuning (Final Model): refined the best raw-band configuration further, from 88.21% to ~89% accuracy.

4.2 What Did Not Improve Performance

- Increasing sample size alone (EXP-01 → EXP-03), without adding temporal or engineered features, did not improve — and slightly reduced — accuracy.
- Adding 28 spectral indices to the 4-season raw-band Random Forest model (EXP-04) left accuracy and Macro F1 unchanged at 87.00% / 0.87, indicating the raw multi-seasonal bands already captured most of the discriminative signal.

5. Final Model — Tuned XGBoost

Based on the complete experimental evaluation, the final model was built on the best-performing configuration — 40 raw multi-seasonal Sentinel-2 spectral features — with XGBoost hyperparameters optimized to maximize performance.

5.1 Final Dataset Configuration

Parameter	Configuration
Task	Multi-class LULC Classification
Dataset Size	90,000 samples
Number of Classes	9
Samples per Class	10,000
Train/Test Split	80/20
Training Samples	72,000
Test Samples	18,000
Temporal Coverage	4 Seasons (Q1–Q4)
Spectral Bands per Season	10
Total Input Features	40

Parameter	Configuration
Spectral Indices	None
Model	XGBoost Classifier
Hyperparameter Tuning	Yes
Random State	42

5.2 Tuned Hyperparameters

Hyperparameter	Final Value
n_estimators	700
max_depth	10
learning_rate	0.05
subsample	0.8
colsample_bytree	0.8
objective	multi:softmax
num_class	9
eval_metric	mlogloss
random_state	42
n_jobs	-1

- 700 estimators, with a learning rate of 0.05, balances model capacity against overfitting through many small incremental updates.
- Max depth of 10 allows trees to capture complex, non-linear relationships between spectral bands and land cover classes.
- Subsample and column-sampling ratios of 0.8 introduce randomness that improves generalization to unseen data.

5.3 Final Model Performance

Metric	Result
Accuracy	0.89 (~89%)
Macro Precision	0.89
Macro Recall	0.89
Macro F1-score	0.89
Weighted Precision	0.89
Weighted Recall	0.89
Weighted F1-score	0.89
Test Samples	18,000

5.4 Class-wise Performance

Class	Precision	Recall	F1-score	Support
0	0.84	0.82	0.83	2,000
1	0.85	0.87	0.86	2,000
2	0.82	0.81	0.82	2,000
3	0.93	0.91	0.92	2,000
4	0.85	0.87	0.86	2,000
5	0.88	0.89	0.88	2,000
6	0.99	0.98	0.99	2,000
7	0.86	0.89	0.87	2,000
8	0.96	0.92	0.94	2,000
Macro Avg	0.89	0.89	0.89	18,000
Weighted Avg	0.89	0.89	0.89	18,000

- Best-performing class: Class 6 (F1 = 0.99) — highly distinguishable spectral characteristics.
- Second-best: Class 8 (F1 = 0.94), followed by Class 3 (F1 = 0.92).
- Most challenging: Class 2 (F1 = 0.82) — likely spectral overlap with neighboring classes.

5.5 Effect of Hyperparameter Tuning

Model	Accuracy	Macro F1
XGBoost — Before Tuning	88.21%	0.88
XGBoost — Tuned (Final)	~89%	0.89

Tuning delivered a modest but consistent gain of approximately +0.7 percentage points in accuracy and +0.01 in Macro F1 over the untuned EXP-04 configuration — a meaningful refinement given the already strong starting baseline.

5.6 Final Model Pipeline

Sentinel-2 Satellite Data

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4 Seasonal Observations (Q1 | Q2 | Q3 | Q4)

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10 Spectral Bands per Season

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40 Input Features

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Train / Test Split (80% / 20%)

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XGBoost Classifier

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Hyperparameter Optimization

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Tuned XGBoost Model

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9-Class LULC Prediction

6. Key Findings

- XGBoost consistently outperformed Random Forest across nearly every experimental configuration in this project.
- Multi-seasonal spectral information was the single most impactful factor for classification accuracy, outweighing both dataset size and spectral-index augmentation.
- Increasing sample size alone, without richer features, produced no meaningful improvement — data volume is not a substitute for feature richness.
- Spectral indices provided benefit in the smaller, 3-season EXP-02 setup, but added no measurable improvement on top of the already-rich 4-season raw-band representation in EXP-04.
- Certain LULC classes (e.g., Class 80 / Class 6 in the final model) were consistently easy to classify across all experiments, indicating strong, stable spectral separability.
- Certain classes (e.g., Class 30 / Class 2) were consistently the most difficult, suggesting persistent spectral overlap with neighboring land cover types that further feature engineering did not fully resolve.
- Hyperparameter tuning provided a final, incremental performance gain, confirming the raw multi-seasonal feature set combined with XGBoost as the strongest overall configuration.

7. Conclusion

This project systematically evaluated the effects of feature engineering, dataset size, and multi-seasonal information on LULC classification performance using Sentinel-2 imagery and ensemble machine learning models. Through four progressive experiments, it was established that multi-seasonal raw spectral information, combined with a sufficiently large and balanced dataset, provides the strongest foundation for accurate classification — outperforming configurations that relied solely on larger sample sizes or additional spectral indices.

The final model, a hyperparameter-tuned XGBoost classifier using 40 raw multi-seasonal Sentinel-2 spectral features, achieved the best result of the entire project: approximately 89% accuracy and a Macro F1-score of

0.89 on 18,000 unseen test samples. This configuration is recommended as the final, deployable solution for nine-class LULC classification within the scope of this project.

8. Future Work

- Explore additional spectral indices or texture-based features specifically targeting the most challenging classes (e.g., Class 2 / Class 30).
- Investigate deep learning approaches (e.g., CNNs or transformer-based models) for potential further performance gains.
- Extend temporal coverage beyond four seasons, or incorporate multi-year data, to evaluate long-term temporal stability.
- Validate the final model on an independent, geographically distinct test region to assess generalization beyond the current dataset.